

## ■ 2.6.8 Advanced Topic: Formatting Functions as Operators

|                                      |                                                           |
|--------------------------------------|-----------------------------------------------------------|
| <code>Prefix[f[x], h]</code>         | prefix form $h \ x$                                       |
| <code>Postfix[f[x], h]</code>        | postfix form $x \ h$                                      |
| <code>Infix[f[x, y, ...], h]</code>  | infix form $x \ h \ y \ h \ \dots$                        |
| <code>Prefix[f[x]]</code>            | standard prefix form $f@x$                                |
| <code>Postfix[f[x]]</code>           | standard postfix form $x//f$                              |
| <code>Infix[f[x, y, ...]]</code>     | standard infix form $x \sim f \sim y \sim f \sim \dots$   |
| <code>PrecedenceForm[expr, n]</code> | an object to be parenthesized with a precedence level $n$ |

Output forms for operators.

This prints with `f` represented by the “prefix operator” `<>`.

```
In[1]:= Prefix[f[x], "<>"]
Out[1]= <>x
```

Here is output with the “infix operator” `==`.

```
In[2]:= s = Infix[{a, b, a/b, b/a}, " == "]
Out[2]= a == b == (a/b) == (b/a)
```

By default, the “infix operator” `==` is assumed to have “higher precedence” than `Power`, so no parentheses are inserted.

```
In[3]:= s^2
Out[3]= a == b == (a/b) == (b/a)^2
```

Here `==` is treated as an operator with precedence 100. This precedence turns out to be low enough that parentheses are inserted.

```
In[4]:= PrecedenceForm[s, 100]^2
Out[4]= (a == b == (a/b) == (b/a))^2
```